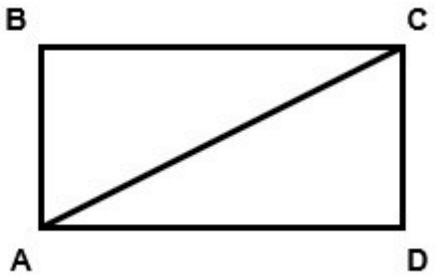
HINDUSTAN
INSTITUTE OF TECHNOLOGY & SCIENCE
(DEEMED TO BE UNIVERSITY)

**EMA51005 – Discrete Mathematics (Question Bank)**

**UNIT-V – Graph Theory**

## Part A

1. Define a pseudograph.
2. What is complete graph? Draw a complete graph on 4 vertices.
3. Draw a graph which contains Eulerian path and circuit.
4. Define a 3 – regular graphs with an example.
5. What you mean by bipartite graph and give an example.
6. Check Hamilton circuit is existing or not for the following graph.

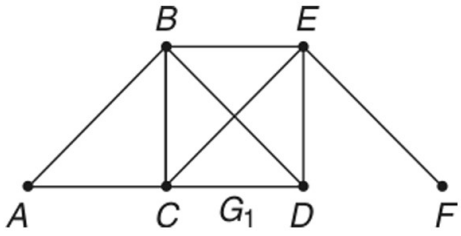


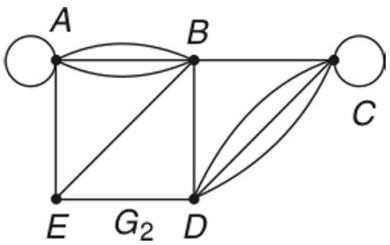

7. State Handshaking theorem.
8. Define a multigraph.
9. Draw a complete bipartite graph $K_{2,3}$.
10. Define a regular graph.

## Part B

1. Prove that the number of vertices of odd degree in an undirected graph is even.
2. Draw a graph which contains,
   (i) an Eulerian circuit that is also a Hamiltonian circuit,
   (ii) an Eulerian circuit, but not a Hamiltonian circuit.
   (iii) A Hamiltonian circuit, but not an Eulerian circuit.
3. State and prove the Handshaking theorem.
4. Write short notes on the following with illustration:
   (i) Adjacency Matrix and its properties.
   (ii) Incident Matrix and its properties
5. Define graph isomorphism and give an example for isomorphism and non-isomorphism with justification.
6. Find the number of vertices, the number of edges and the degree of each vertex in the following undirected graphs $G_1$and $G_2$. Verify also the handshaking theorem in each case.

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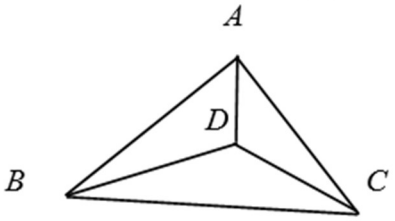


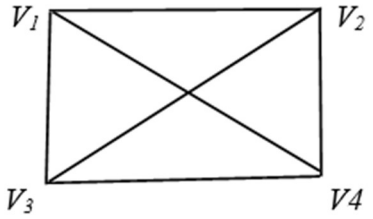

7. Draw the graph with 5 vertices, A, B, C, D, E such that deg (A) =3, B is an odd vertex, deg(C) = 2 and D and E are adjacent.
8. Prove that the number of edges in a bipartite graph with *n* vertices is atmost ($n^2/2$).
9. Explain the Eulerian Graph of Eulerian path, Eulerian Circuit and also the Hamiltonian Graph of Hamiltonian path, Hamiltonian Circuit with suitable examples of graphical representations.
10. Write a few properties of Hamiltonian Circuit.

## Part C

1. Prove that the maximum number of edges in a simple disconnected graph with '$n$' vertices and k components is $(n-k)(n-k+1)/2$
2. (i) Define isomorphism of a graph *G*.
   (ii) Using circuits, check whether the following pairs of graphs $G_1$ and $G_2$ are isomorphic or not and justify your answer.

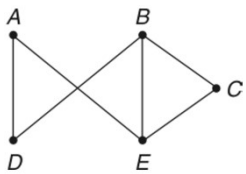



$G_1$

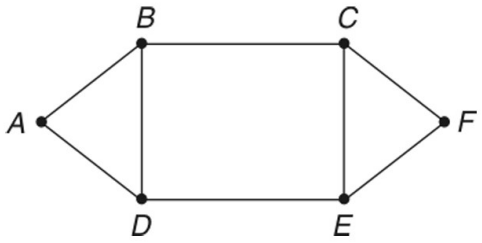



$G_2$

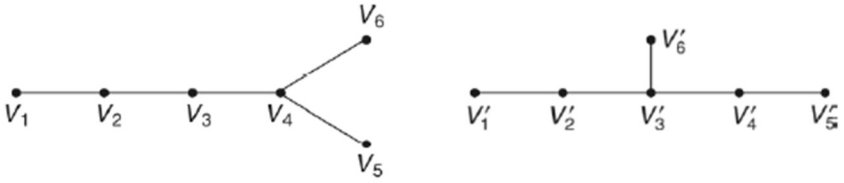
3. (i) If *G= (V, E)* is an undirected graph with *e* edges, then prove that $\sum_i \deg(v_i) = 2e$.
   (ii) Prove that the number of vertices of odd degree in an undirected graph is even
4. If all the vertices of an undirected graph are each of odd degree *k*, show that the number of edges of the graph is a multiple of *k*.
5. (i) Find which of the following vertex sequences are simple paths, paths, closed paths (circuits) and simple circuits with respect to the graph:
   (a) A – D – E – B – C
   (b) A – D – B – C – E
   (c) A – E – C – B – E – A
   (d) C – B – D – A – E – C
   (e) A – D – B – E – C – B




   (ii) Find all the simple paths from A to F and all the circuits in the graph given:




6. Prove that a simple graph with $n$ vertices must be connected if it has more than $(n-1)(n-2)/2$.
7. Explain the path, cycles and connectivity with suitable example of graphical representations.
8. For each of the following degree sequences, find if there exists a graph. In each case, either draw a graph or explain why no graphs exists.
   (i) (4,4,4,3,2)
   (ii) (5,5,4,3,2,1)
   (iii)(3,3,3,3,2)
   (iv)(3,3,3,3,3)
9. Determine whether the graph shown in the following figures are isomorphic.




(a) (b)

10. Determine whether the graph shown in the following figures are isomorphic.

